MSG No. 72.547 $I_b bam$ [Type IV, orthorhombic]

Table 1: Wyckoff site: `8a`, site symmetry: `222`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,3,4]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[9,10,11,12]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[13,14,15,16]`
5	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[17,18,19,20]`
6	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[21,22,23,24]`
7	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[25,26,27,28]`
8	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[29,30,31,32]`

Table 2: Wyckoff site: `8b`, site symmetry: `2'22'`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	`[1,3,18,20]`
2	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[2,4,17,19]`
3	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	`[5,7,22,24]`
4	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[6,8,21,23]`
5	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[9,11,26,28]`
6	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[10,12,25,27]`
7	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	`[13,15,30,32]`
8	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[14,16,29,31]`

Table 3: Wyckoff site: `8c`, site symmetry: `..2/m`

No.	position	mapping
1	$[0, 0, 0]$	`[1,4,5,8]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[2,3,6,7]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,12,13,16]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[10,11,14,15]`
5	$\left[0, \frac{1}{2}, 0\right]$	`[17,20,21,24]`
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[18,19,22,23]`
7	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[25,28,29,32]`
8	$\left[\frac{1}{2}, 0, 0\right]$	`[26,27,30,31]`

Table 4: Wyckoff site: `8d`, site symmetry: `..2'/m`

No.	position	mapping
1	$\left[0, \frac{1}{4}, 0\right]$	`[1,8,20,21]`
2	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[2,7,19,22]`
3	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[3,6,18,23]`
4	$\left[0, \frac{3}{4}, 0\right]$	`[4,5,17,24]`
5	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right]$	`[9,16,28,29]`

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[10,15,27,30]
7	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[11,14,26,31]
8	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\right]$	[12,13,25,32]

Table 5: Wyckoff site: 8e, site symmetry: 2/m’..

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	[1,2,29,30]
2	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[3,4,31,32]
3	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	[5,6,25,26]
4	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[7,8,27,28]
5	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[9,10,21,22]
6	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[11,12,23,24]
7	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[13,14,17,18]
8	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[15,16,19,20]

Table 6: Wyckoff site: 8f, site symmetry: 2’/m’..

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,13,18,30]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[2,14,17,29]
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[3,15,20,32]
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[4,16,19,31]
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[5,9,22,26]
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[6,10,21,25]
7	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[7,11,24,28]
8	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[8,12,23,27]

Table 7: Wyckoff site: 8g, site symmetry: m’2’m

No.	position	mapping
1	$\left[\frac{1}{4}, y, 0\right]$	[1,8,27,30]
2	$\left[\frac{1}{4}, -y, \frac{1}{2}\right]$	[2,7,28,29]
3	$\left[\frac{3}{4}, y, \frac{1}{2}\right]$	[3,6,25,32]
4	$\left[\frac{3}{4}, -y, 0\right]$	[4,5,26,31]
5	$\left[\frac{3}{4}, y+\frac{1}{2}, \frac{1}{2}\right]$	[9,16,19,22]
6	$\left[\frac{3}{4}, \frac{1}{2}-y, 0\right]$	[10,15,20,21]
7	$\left[\frac{1}{4}, y+\frac{1}{2}, 0\right]$	[11,14,17,24]
8	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{1}{2}\right]$	[12,13,18,23]

Table 8: Wyckoff site: `16h`, site symmetry: `..2`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4]`
2	$[0,\ 0,\ \frac{1}{2}-z]$	`[2,3]`
3	$[0,\ 0,\ -z]$	`[5,8]`
4	$[0,\ 0,\ z+\frac{1}{2}]$	`[6,7]`
5	$[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}]$	`[9,12]`
6	$[\frac{1}{2},\ \frac{1}{2},\ -z]$	`[10,11]`
7	$[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z]$	`[13,16]`
8	$[\frac{1}{2},\ \frac{1}{2},\ z]$	`[14,15]`
9	$[0,\ \frac{1}{2},\ z]$	`[17,20]`
10	$[0,\ \frac{1}{2},\ \frac{1}{2}-z]$	`[18,19]`
11	$[0,\ \frac{1}{2},\ -z]$	`[21,24]`
12	$[0,\ \frac{1}{2},\ z+\frac{1}{2}]$	`[22,23]`
13	$[\frac{1}{2},\ 0,\ z+\frac{1}{2}]$	`[25,28]`
14	$[\frac{1}{2},\ 0,\ -z]$	`[26,27]`
15	$[\frac{1}{2},\ 0,\ \frac{1}{2}-z]$	`[29,32]`
16	$[\frac{1}{2},\ 0,\ z]$	`[30,31]`

Table 9: Wyckoff site: `16i`, site symmetry: `..2'`

No.	position	mapping
1	$[0,\ \frac{1}{4},\ z]$	`[1,20]`
2	$[0,\ \frac{3}{4},\ \frac{1}{2}-z]$	`[2,19]`
3	$[0,\ \frac{1}{4},\ \frac{1}{2}-z]$	`[3,18]`
4	$[0,\ \frac{3}{4},\ z]$	`[4,17]`
5	$[0,\ \frac{3}{4},\ -z]$	`[5,24]`
6	$[0,\ \frac{1}{4},\ z+\frac{1}{2}]$	`[6,23]`
7	$[0,\ \frac{3}{4},\ z+\frac{1}{2}]$	`[7,22]`
8	$[0,\ \frac{1}{4},\ -z]$	`[8,21]`
9	$[\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}]$	`[9,28]`
10	$[\frac{1}{2},\ \frac{1}{4},\ -z]$	`[10,27]`
11	$[\frac{1}{2},\ \frac{3}{4},\ -z]$	`[11,26]`
12	$[\frac{1}{2},\ \frac{1}{4},\ z+\frac{1}{2}]$	`[12,25]`
13	$[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-z]$	`[13,32]`
14	$[\frac{1}{2},\ \frac{3}{4},\ z]$	`[14,31]`
15	$[\frac{1}{2},\ \frac{1}{4},\ z]$	`[15,30]`
16	$[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}-z]$	`[16,29]`

Table 10: Wyckoff site: `16j`, site symmetry: `2..`

No.	position	mapping
1	$[x,\ 0,\ \frac{1}{4}]$	`[1,2]`
2	$[-x,\ 0,\ \frac{1}{4}]$	`[3,4]`
3	$[-x,\ 0,\ \frac{3}{4}]$	`[5,6]`
4	$[x,\ 0,\ \frac{3}{4}]$	`[7,8]`

continued ...

Table 10

No.	position	mapping
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[9,10]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[11,12]
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[13,14]
8	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[15,16]
9	$\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[17,18]
10	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[19,20]
11	$\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[21,22]
12	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[23,24]
13	$\left[x+\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	[25,26]
14	$\left[\frac{1}{2}-x,\ 0,\ \frac{3}{4}\right]$	[27,28]
15	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{4}\right]$	[29,30]
16	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[31,32]

Table 11: Wyckoff site: 16k, site symmetry: 2'..

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[1,18]
2	$\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[2,17]
3	$\left[-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[3,20]
4	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[4,19]
5	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[5,22]
6	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[6,21]
7	$\left[x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[7,24]
8	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[8,23]
9	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	[9,26]
10	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	[10,25]
11	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[11,28]
12	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[12,27]
13	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[13,30]
14	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[14,29]
15	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{4}\right]$	[15,32]
16	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	[16,31]

Table 12: Wyckoff site: 16l, site symmetry: .2.

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	[1,3]
2	$\left[0,\ -y,\ \frac{1}{4}\right]$	[2,4]
3	$\left[0,\ -y,\ \frac{3}{4}\right]$	[5,7]
4	$\left[0,\ y,\ \frac{3}{4}\right]$	[6,8]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[9,11]
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[10,12]
7	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[13,15]
8	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[14,16]

continued ...

Table 12

No.	position	mapping
9	$\left[0,\, y+\frac{1}{2},\, \frac{1}{4}\right]$	[17,19]
10	$\left[0,\, \frac{1}{2}-y,\, \frac{1}{4}\right]$	[18,20]
11	$\left[0,\, \frac{1}{2}-y,\, \frac{3}{4}\right]$	[21,23]
12	$\left[0,\, y+\frac{1}{2},\, \frac{3}{4}\right]$	[22,24]
13	$\left[\frac{1}{2},\, y,\, \frac{3}{4}\right]$	[25,27]
14	$\left[\frac{1}{2},\, -y,\, \frac{3}{4}\right]$	[26,28]
15	$\left[\frac{1}{2},\, -y,\, \frac{1}{4}\right]$	[29,31]
16	$\left[\frac{1}{2},\, y,\, \frac{1}{4}\right]$	[30,32]

Table 13: Wyckoff site: 16m, site symmetry: ..m

No.	position	mapping
1	$[x,\, y,\, 0]$	[1,8]
2	$\left[x,\, -y,\, \frac{1}{2}\right]$	[2,7]
3	$\left[-x,\, y,\, \frac{1}{2}\right]$	[3,6]
4	$[-x,\, -y,\, 0]$	[4,5]
5	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[9,16]
6	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, 0\right]$	[10,15]
7	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, 0\right]$	[11,14]
8	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[12,13]
9	$\left[x,\, y+\frac{1}{2},\, 0\right]$	[17,24]
10	$\left[x,\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[18,23]
11	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[19,22]
12	$\left[-x,\, \frac{1}{2}-y,\, 0\right]$	[20,21]
13	$\left[x+\frac{1}{2},\, y,\, \frac{1}{2}\right]$	[25,32]
14	$\left[x+\frac{1}{2},\, -y,\, 0\right]$	[26,31]
15	$\left[\frac{1}{2}-x,\, y,\, 0\right]$	[27,30]
16	$\left[\frac{1}{2}-x,\, -y,\, \frac{1}{2}\right]$	[28,29]

Table 14: Wyckoff site: 16n, site symmetry: m'..

No.	position	mapping
1	$\left[\frac{1}{4},\, y,\, z\right]$	[1,30]
2	$\left[\frac{1}{4},\, -y,\, \frac{1}{2}-z\right]$	[2,29]
3	$\left[\frac{3}{4},\, y,\, \frac{1}{2}-z\right]$	[3,32]
4	$\left[\frac{3}{4},\, -y,\, z\right]$	[4,31]
5	$\left[\frac{3}{4},\, -y,\, -z\right]$	[5,26]
6	$\left[\frac{3}{4},\, y,\, z+\frac{1}{2}\right]$	[6,25]
7	$\left[\frac{1}{4},\, -y,\, z+\frac{1}{2}\right]$	[7,28]
8	$\left[\frac{1}{4},\, y,\, -z\right]$	[8,27]
9	$\left[\frac{3}{4},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9,22]
10	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, -z\right]$	[10,21]
11	$\left[\frac{1}{4},\, y+\frac{1}{2},\, -z\right]$	[11,24]
12	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[12,23]

continued ...

Table 14

No.	position	mapping
13	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[13,18]
14	$\left[\frac{1}{4},\ y+\frac{1}{2},\ z\right]$	[14,17]
15	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ z\right]$	[15,20]
16	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[16,19]

Table 15: Wyckoff site: 32o, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[2]
3	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	[6]
7	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	[7]
8	$[x,\ y,\ -z]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9]
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[10]
11	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[14]
15	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[15]
16	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[16]
17	$\left[x,\ y+\frac{1}{2},\ z\right]$	[17]
18	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[18]
19	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[19]
20	$\left[-x,\ \frac{1}{2}-y,\ z\right]$	[20]
21	$\left[-x,\ \frac{1}{2}-y,\ -z\right]$	[21]
22	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[22]
23	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[23]
24	$\left[x,\ y+\frac{1}{2},\ -z\right]$	[24]
25	$\left[x+\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[25]
26	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	[26]
27	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	[27]
28	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[28]
29	$\left[\frac{1}{2}-x,\ -y,\ \frac{1}{2}-z\right]$	[29]
30	$\left[\frac{1}{2}-x,\ y,\ z\right]$	[30]
31	$\left[x+\frac{1}{2},\ -y,\ z\right]$	[31]
32	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	[32]